



Linear equations in one unknown: Orientation map

An equation constrains an unknown

Subject: Mathematics

Topic: Linear equations

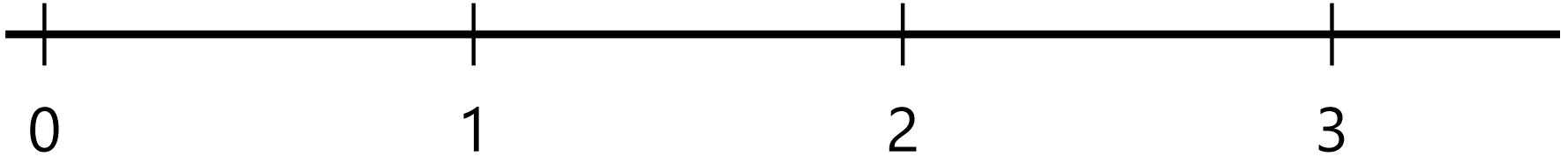
Concept: An equation constrains an unknown

Concept difficulty: Medium concept

Hard transition

Read $a \cdot x + b = c$ as a statement that is true for some values of the unknown and false for others, rather than as an instruction to work out a left-hand side.

Domain: the rationals · x (dimensionless)



Draw the rationals as one labelled axis with integer ticks before anything is marked.

Stage 1 of 3

Place the exact solution between its bracketing integers so that it is visibly not equal to either.

Governing relation

$$x = \frac{c - b}{a}$$

Subtracting b and dividing by a non-zero a isolates the unknown using reversible operations, so the solution set is unchanged.

- a is non-zero; if $a = 0$ the statement is either never true or always true.
- Both operations are reversible, which is why the solution set is preserved.

- The declared domain is the rationals.

Compact anchor

Is $x = 2$ a solution of $3x + 2 = 9$?

No: $3 \times 2 + 2 = 8$, and $8 \neq 9$, so the claim is false for $x = 2$.

Before you go on

Does $x = 2$ satisfy $3x + 2 = 9$? Show how you decided.

Operations that preserve the solution set

Subject: Mathematics

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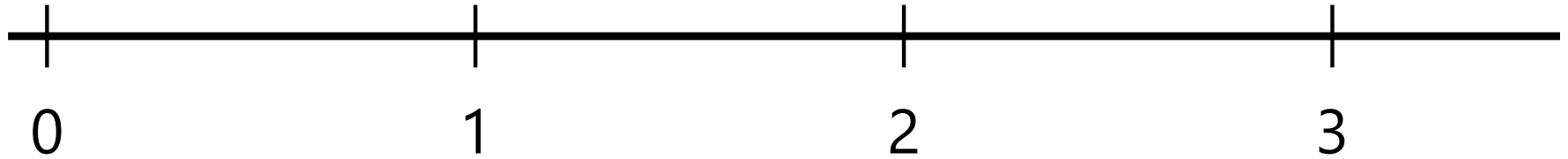
Concept: Operations that preserve the solution set

Concept difficulty: Medium concept

Hard transition

See each manipulation as an operation applied to both sides whose reversibility is what guarantees no solution is gained or lost.

Domain: the rationals · x (dimensionless)



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Compact anchor

Solve $4x + 1 = 13$ using only steps that can be undone.

Subtract 1 from both sides: $4x = 12$. Divide both sides by 4, which is not zero: $x = 3$. Check: $4 \times 3 + 1 = 13$.

Before you go on

Solve $5x - 4 = 11$, naming the operation used at each step and why it is reversible.

An exact solution is not its decimal approximation

Subject: Mathematics

Topic: Linear equations

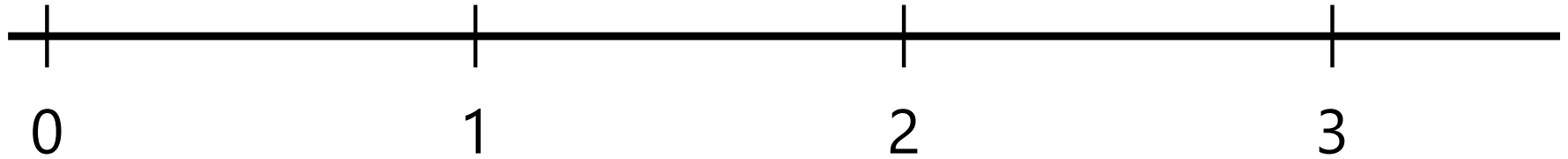
Concept: An exact solution is not its decimal approximation

Concept difficulty: Medium concept

Hard transition

Treat the exact rational and its truncated decimal as two different numbers, and use substitution to show the difference rather than asserting it.

Domain: the rationals · x (dimensionless)



Draw the rationals as one labelled axis with integer ticks before anything is marked.

Stage 1 of 3

Place the exact solution between its bracketing integers so that it is visibly not equal to either.

Governing relation

x exact $\Rightarrow a \cdot x + b = c$ holds exactly

For a nonzero linear coefficient, the residual is a times the error in the candidate value. An exact terminating decimal is an exact rational too; only a changed value introduces a nonzero residual.

- The linear coefficient a is non-zero and all arithmetic is exact over the rationals.
- The decimal approximation differs numerically from the exact solution.

- Rounding is a presentation choice made after the mathematics, never during it.

Compact anchor

Which solves $3x = 2$ exactly: $2/3$ or 0.67 ?

$2/3$, because $3 \times 2/3 = 2$ exactly; $3 \times 0.67 = 2.01$, so 0.67 is a nearby number, not the solution.

Before you go on

Solve $3x + 2 = 9$ over the rationals, and explain why writing the answer as 2.33 would be a different claim.